FYUGP B.Sc. PHYSICS HONOURS

PROPERTIES OF SOLIDS

Semester V · Major Elective (Materials Science)

MODULE III: SEMICONDUCTORS

Past questions

Section A: 1 to 3 marks

Intrinsic and extrinsic semiconductors

1 Explain the concept of holes in semiconductors. **(1 weightage)** *November 2019 (M.Sc.)*

2 What do you meant by indirect band gap semiconductors ? **(1 weightage)** *November 2023 (M.Sc.)*

3 What is a semiconducting material ? Give the significance. **(2 marks)** *March 2025*

Drift velocity, mobility and conductivity

4 An electric field of 100 V/m is applied to a n-type semiconductor having hall co-efficient 0.0125 m^3/C. Determine the current density in the sample if the electron mobility is 0.36 $m^2\ v^{-1}\ s^{-1}$. **(3 weightage)** *November 2022 (M.Sc.)*

5 The mobility of hole is $\mu_h = 0.025\ m^2/Vsec$. What would be the resistivity of p-type silicon if the hall co-efficient of the sample is $2.25 \times 10^{-5}\ m^3/C$? **(3 weightage)** *November 2023 (M.Sc.)*

Carrier concentration and Fermi level

6 Prove that the Fermi level in an intrinsic semiconductor lies in the middle of the forbidden gap if the effective masses of electrons and holes are equal. **(3 weightage)** *November 2019 (M.Sc.)*

Section B: 4 to 5 marks

Carrier concentration and Fermi level

7 Obtain the expressions for the carrier concentrations in intrinsic semiconductor. **(5 weightage)** *November 2022 (M.Sc.)*